

# Test worksheet

## Lesson 1

1. The functions  $f$  and  $fg$  are defined by  $f(x) = e^{(x^2 + 3)}$  for  $x < 0$  and  $fg(x) = e^{(2x)}$  for  $x > 3/2$ . Explain why  $f^{-1}$  exists. [1 mark]

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2. Find an expression for  $f^{-1}(x)$  and state the domain and range of  $f^{-1}$ . [5 marks]

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3. Hence find and simplify an expression for  $g(x)$ . [2 marks]

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